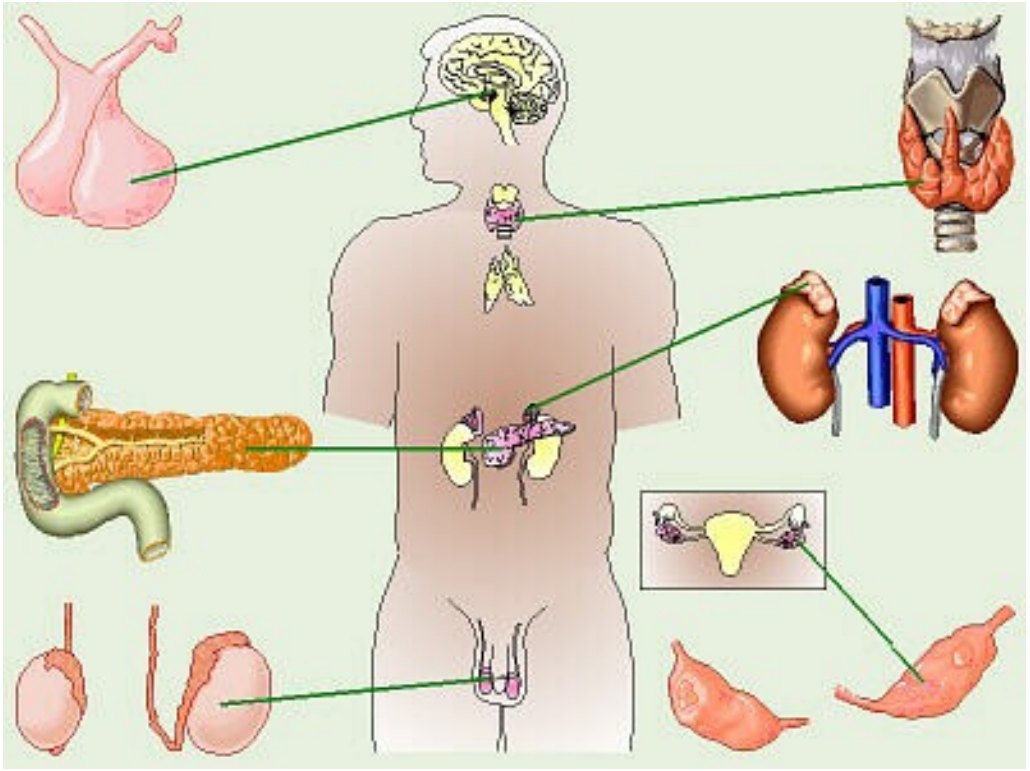
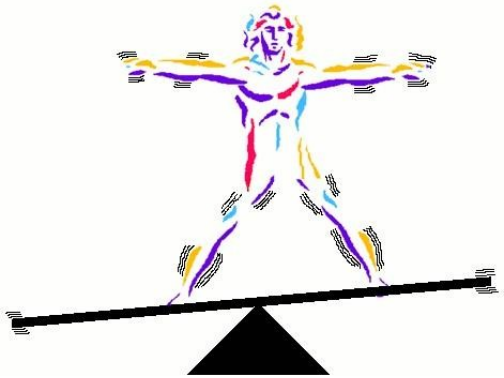
Intro to Endocrine System and Hormones



- The body must maintain internal conditions (homeostasis)
  - Coordination of systems and organs is required to maintain health
    - The endocrine system is one aspect of this coordination
    - Messages must be sent between organs and systems so coordination can happen
      - **Sensor:** detects change to conditions and triggers messenger
      - **Messenger**: sends information to control centre/modulator
      - **Control Centre/Modulator**: processes information and triggers messenger
      - **Messenger**: takes instructions to effector
      - **Effector:** carries out instructions

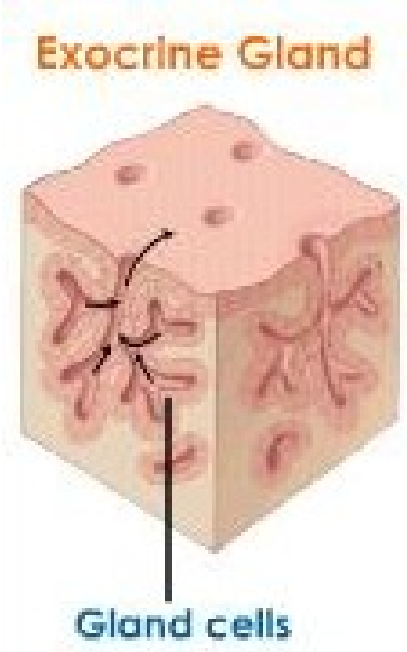


  - Feedback mechanisms allow processes to stop when homeostasis has been reached
  - Maintains balanced internal environment

Glands – Exocrine and Endocrine

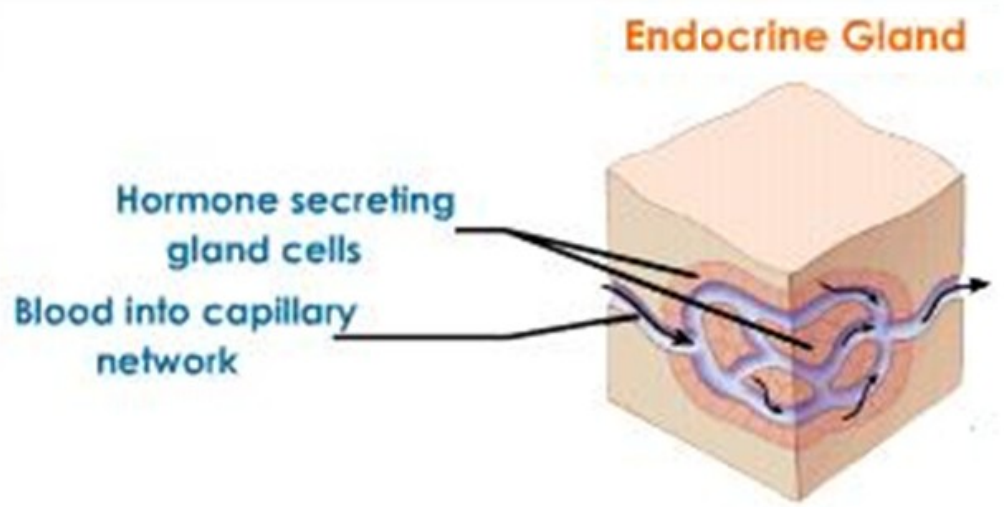
Exocrine glands:

- Eg: sweat glands, mucous glands, salivary glands
- Secretion travels down duct to local area eg. body surface, stomach.
- Do not secrete hormones.

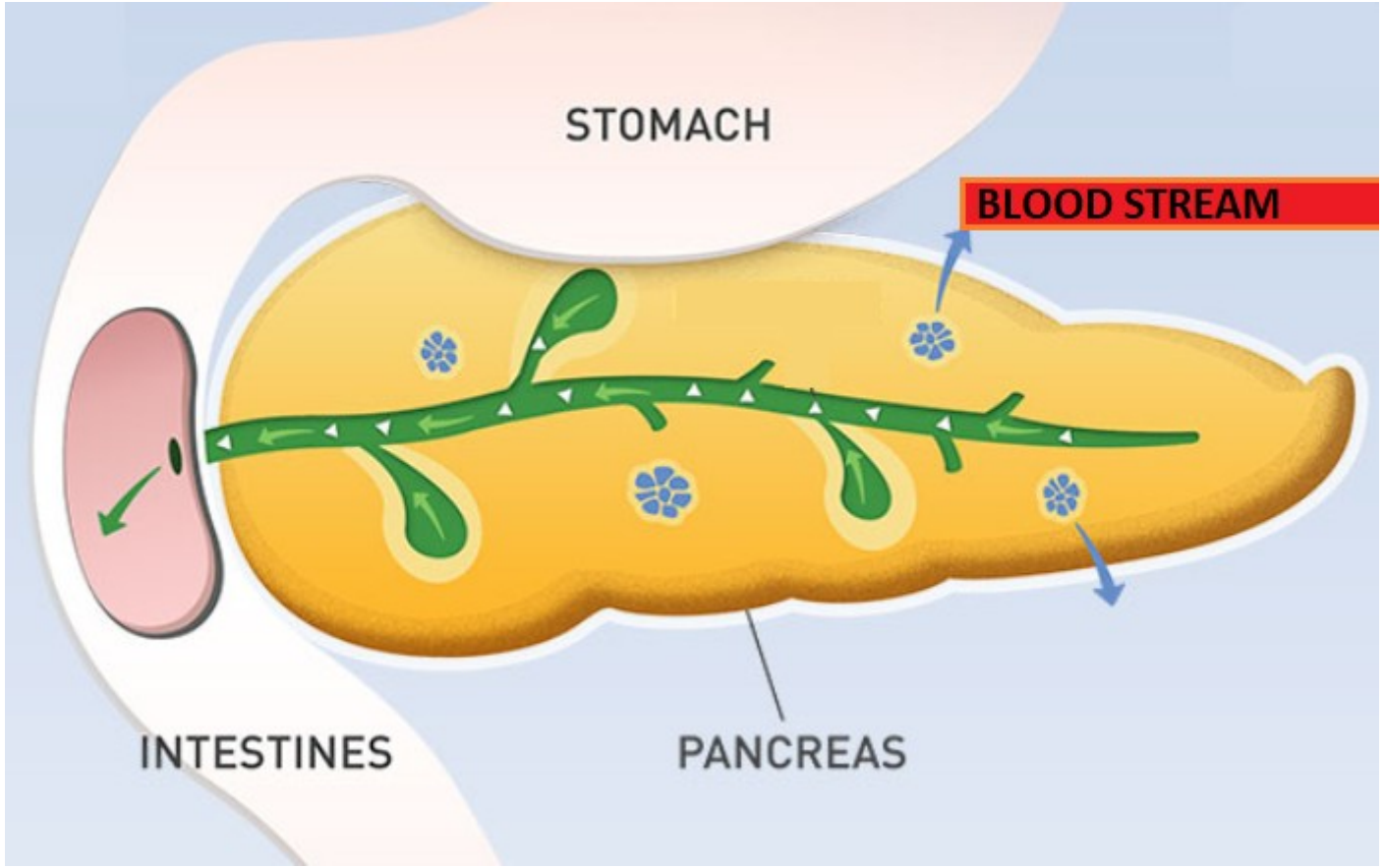


Endocrine glands:

- Eg: pituitary, pineal, pancreas…
- Secrete ***hormones*** into fluid surrounding cells, which then enters the blood stream.
- No ducts



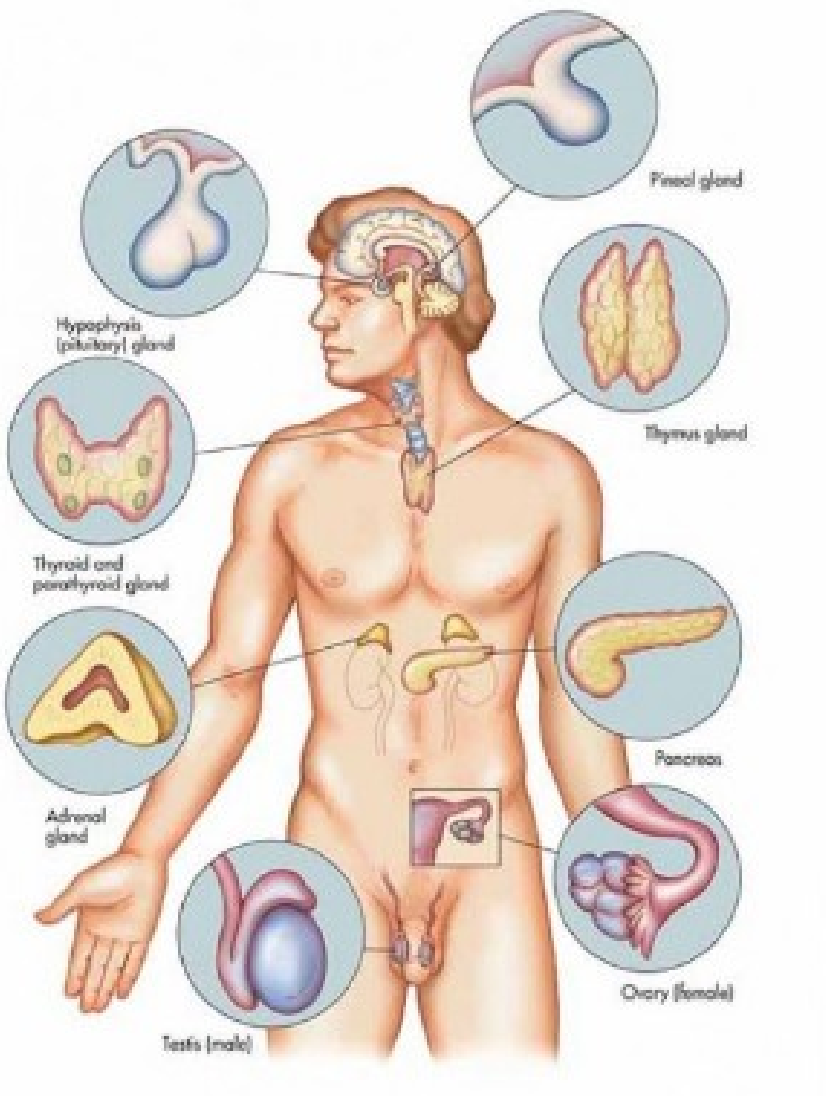
Eg: The pancreas has both exocrine and endocrine functions



- The *exocrine pancreas* is shown here in green and yellow. The tissue shown in yellow is exocrine gland tissue. It produces enzyme secretions that travel down ducts (shown in green) into the intestine.

- The *endocrine pancreas* is shown here in blue. Clusters of endocrine tissue produce hormones (insulin and glucagon) and secrete these directly into the blood stream via surrounding capillaries.

Endocrine System Overview



Structure:

- A series of glands that secrete hormones into the bloodstream.
- Different glands secrete different hormones.

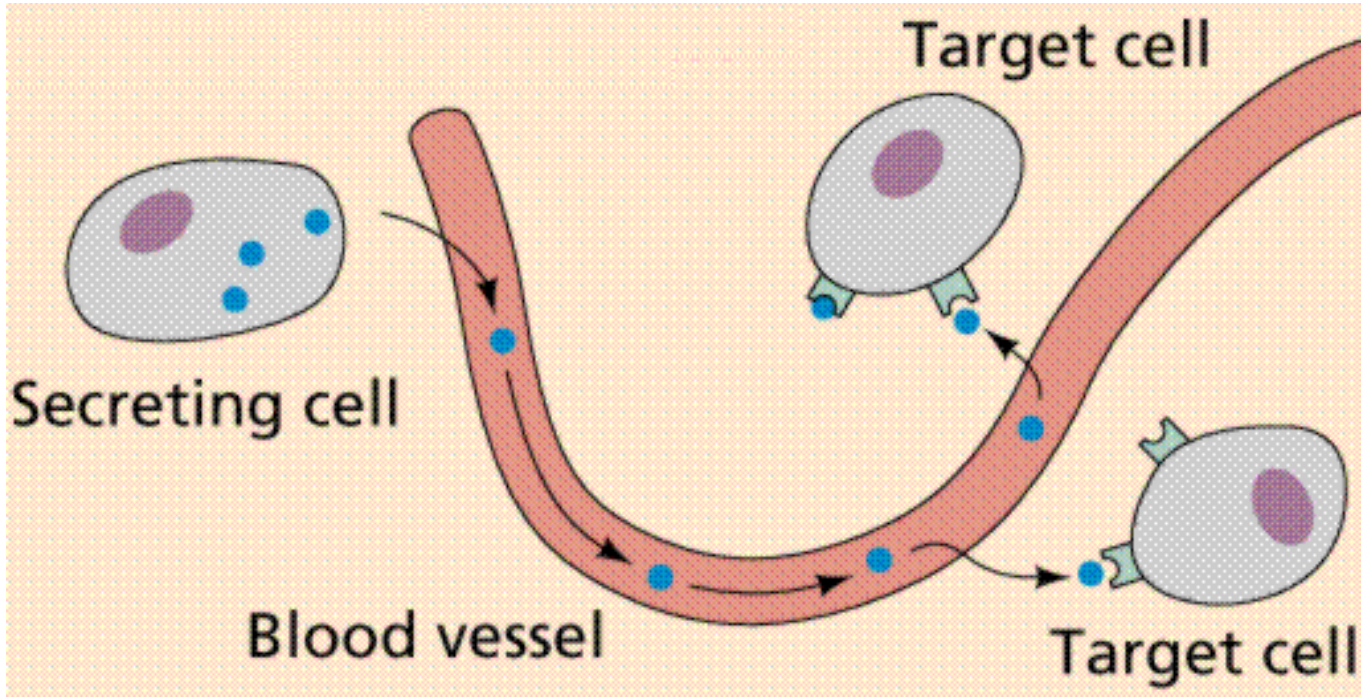
Function:

- To maintain the internal conditions required for health by:
  - Detecting changing internal conditions
  - Producing hormones that affect body cells
  - Releasing these hormones as chemical messengers that enter cells, changing their function to restore internal conditions.

Learning Aim: Describe the function of the endocrine system

What are hormones?

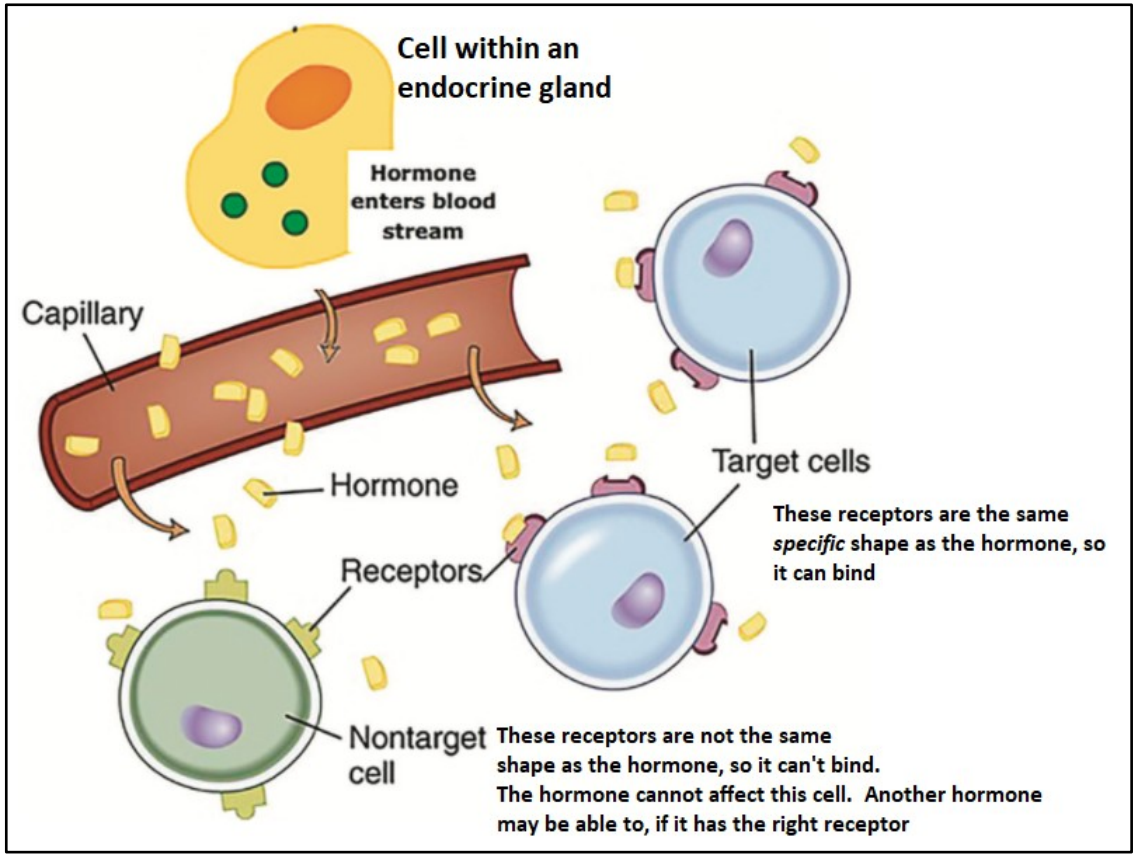
- Are chemical messengers secreted from endocrine glands into the blood.
- Hormones interact with *receptors* on or in cells and affect cell function.



Learning Aim: Define hormones

Hormone Receptors

- Receptors are proteins on or in cells, to which hormones can bind.
- Receptors are *specific* – each type of receptor can only bind with one type of hormone.
  - Some hormones affect all cells, because all cells have receptors for them.
  - Some hormones only affect specific organs (*target organs)* or groups of cells *(target cells)*, because only they have the correct receptors.
  - Different types of cells have different types and numbers of receptors. This is why there is variation in how and where different types of hormones act in the body.

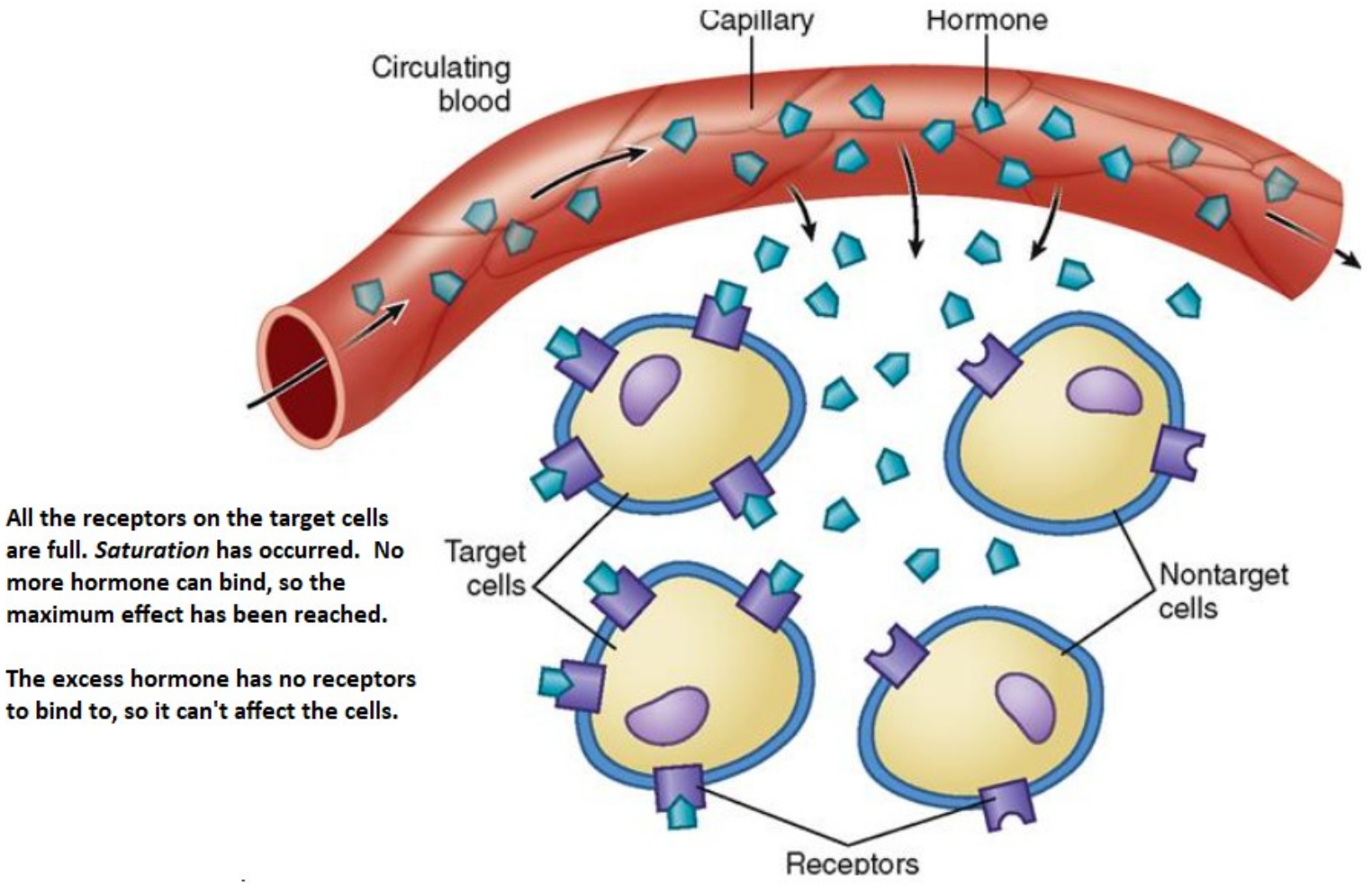


Learning Aim: Explain how hormones interact with cells and which cells may be affected.

Hormone Saturation

- Occurs when all receptor sites are filled, or *saturated*

- At this point, the excess hormone can't have any further effect because there is nowhere for it to bind.



All the receptors on the target cells are full. *Saturation* has occurred. No more hormone can bind, so the maximum effect has been reached.

The excess hormone has no receptors to bind to, so it can't affect the cells.

Learning Aim: Describe hormone saturation

Target cell response to hormones

- Different hormones can act in different ways within the target cell to change the *type, activity or quantity of proteins* produced.
- They can:
  - Activate genes in the nucleus to produce specific enzymes or structural proteins
  - Change the shape or structure of enzymes to turn them ‘on’ or ‘off’
  - Change the rate of transcription and translation to change rate of protein production.

Example: Growth hormone increases the production of proteins involved in bone and muscle development.

Hormone clearance

- Removal of hormones from action, by breaking them down.
- This can happen:
  - In the target cells (sometimes)
  - In the liver and kidneys (mostly)
- Excreted in bile and urine.

Learning Aim: Describe ways in which target cells respond to hormones, and how hormones are cleared from the body

Control of hormone secretions

- Hormone action must be regulated so that stable internal conditions are maintained.
- Over-secretion or under-secretion of a hormone will cause the body to function abnormally.
- To maintain homeostasis, hormone production must be “switched off” once the hormone has done it’s job.
- This is regulated by *negative feedback* systems:
    - Hormone levels are detected by the endocrine gland, which then stops producing that hormone when levels are normal.
    - Sometimes the hypothalamus is involved and produces inhibiting factors once the hormone has done its job.

Learning Aim: Briefly explain how hormone secretions are controlled.

Example of negative feedback

